

Statistics_project

Stephen Lynch & Abigail Sandquist

Appendix

This project investigates how age and years of experience affect employee salary using a dataset containing age, gender, job title, years of experience, and salary. The analysis examines each of the research questions outlined in the proposal: the strength of the relationship between years of experience and salary, whether age contributes additional explanatory power beyond experience, whether salary patterns differ across genders or job titles, the rate at which salary increases with additional experience, and whether salary can be accurately predicted using statistical models. Using descriptive statistics, data visualizations, and linear regression models, we evaluate how these variables interact and identify the most influential predictors of earnings. The findings aim to clarify how professional growth impacts salary progression and provide insight into broader trends in compensation across demographic and job-related categories. This study contributes to understanding the key determinants of salary and supports future work in salary prediction and workforce analytics.

Introduction

Understanding what drives differences in employee salaries is an important question in both workforce analytics and applied statistics. Salary is influenced by many factors, but two of the most commonly discussed predictors are age and years of experience. As individuals gain professional experience, they typically acquire new skills, demonstrate increased responsibility, and advance within their roles—all of which are expected to contribute to higher earnings. Age may also play a role, as it often reflects career stage, career longevity, and time spent in the labor force. However, the degree to which age and experience independently or jointly influence salary remains an empirical question and can vary across industries, job titles, and demographic groups. This research is significant because salary growth is closely tied to career development, economic mobility, and long-term financial stability. Employers use salary models to make compensation decisions, while individuals rely on salary expectations to make choices about education, job changes, and skill development. Understanding how much experience contributes to salary, how strongly age correlates with earnings, and whether demographic or job-related disparities exist can provide valuable insight for employees, employers,

policymakers, and workforce researchers. Despite the importance of these issues, salary data is often complex and influenced by multiple interacting variables. This project uses statistical methods to analyze a data set containing age, gender, job title, years of experience, and salary to better understand the relationships among these variables. Specifically, we investigate how salary changes with increasing experience, whether age adds predictive value beyond experience, whether salary differs across genders or job titles, and whether statistical models can accurately predict salary based on these factors. By examining these relationships, this research aims to deepen our understanding of the key determinants of salary and to highlight potential differences or inequalities within the workforce. The results contribute to broader discussions about compensation fairness, career progression, and the factors that most strongly influence earnings in professional environments.

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    4.0.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
df <- read_csv("Salary_Data.csv")
```

```
Rows: 6704 Columns: 6
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (3): Gender, Education Level, Job Title
```

```
dbl (3): Age, Years of Experience, Salary
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df <- df %>%
  mutate(
    `Education Level` = `Education Level` %>%
      str_remove(" Degree")) %>%
  subset(`Education Level` != "NA")

df[5876,3] <- "PhD"
```

The dataset is imported using `read_csv()`, which reads the raw salary information into R as a structured data frame. This step establishes the foundation for all subsequent analysis by loading the variables needed for the study, including age, gender, education level, job title, experience, and salary. Once imported, the data can be inspected, cleaned, and transformed to ensure accuracy and consistency before modeling. This section performs essential data cleaning to standardize the “Education Level” variable and remove invalid entries. The string manipulation ensures that education labels are concise and consistent, improving interpretability in later statistical models. Additionally, an incorrectly coded entry is manually corrected to “PhD” to preserve the accuracy of the dataset. These steps help maintain data quality and reduce potential bias in the analysis.

Question 1

How does the number of years of experience impact salary levels?

```
salary_years_exp <- lm(data = df, Salary ~ `Years of Experience`)
summary(salary_years_exp)
```

Call:

```
lm(formula = Salary ~ `Years of Experience`, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-148237	-22378	-5565	21014	100575

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	58284.69	632.79	92.11	<2e-16 ***
`Years of Experience`	7046.73	62.58	112.61	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31030 on 6696 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.6544, Adjusted R-squared: 0.6544

F-statistic: 1.268e+04 on 1 and 6696 DF, p-value: < 2.2e-16

Model Description: A simple linear regression examining salary as a function of years of experience.

Key Findings:

Coefficient for Experience: ~\$7,046 per year Each additional year of experience corresponds, on average, to a statistically significant and substantial increase in salary.

R^2 0.65 The model explains 65% of variation in salary using experience alone—surprisingly strong for a single variable.

Interpretation: Experience is a dominant predictor of salary. However, 35% of salary variation remains unexplained, suggesting the influence of additional factors such as education, job title, or industry.

Conclusion: Yes, Years of Experience is one of the most important factors when it comes to modeling salary. It accounts for 65% of the variation by itself without the addition of other predictor variables.

Question 2

Does age have an independent effect on salary apart from years of experience?

```
df$Age_YoE <- df$Age * df$`Years of Experience`  
  
lm2 <- lm(data = df, Salary ~ `Years of Experience` + `Age` + Age_YoE)  
  
summary(lm2)
```

Call:

```
lm(formula = Salary ~ `Years of Experience` + Age + Age_YoE,  
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-112358	-20384	-5970	14445	105329

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	25609.612	3718.041	6.888	6.18e-12 ***
`Years of Experience`	18692.213	299.977	62.312	< 2e-16 ***
Age	479.209	141.472	3.387	0.00071 ***
Age_YoE	-246.339	6.499	-37.906	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27850 on 6694 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.7219, Adjusted R-squared: 0.7217

F-statistic: 5791 on 3 and 6694 DF, p-value: < 2.2e-16

An interaction term is created to examine whether the combined effect of age and experience contributes additional explanatory power in predicting salary. This term allows the regression model to capture how salary growth may differ not only by age or experience individually, but by the interaction of the two—reflecting how time in the workforce and life stage might jointly impact earnings.

Key Findings:

Experience coefficient jumps significantly (~18k), meaning that when controlling for age, experience becomes even more impactful.

Age remains statistically significant, but its direct effect is relatively small (~\$479 per year).

Interaction term (Age × Experience) is negative, large, and significant. Interpretation: the value of experience declines with age, possibly because experience gained later in life may be valued differently by employers than early-career experience.

R² 0.72, an improvement over the simple linear model, showing that age and the interaction help explain more of salary variation.

```
aov_model <- aov(Salary ~ `Years of Experience` + `Age` + Age_YoE , data = df)
summary(aov_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
`Years of Experience`	1	1.221e+13	1.221e+13	15750.0	<2e-16 ***
Age	1	1.441e+11	1.441e+11	185.9	<2e-16 ***
Age_YoE	1	1.114e+12	1.114e+12	1436.9	<2e-16 ***
Residuals	6694	5.191e+12	7.755e+08		

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
3 observations deleted due to missingness
```

An ANOVA (analysis of variance) is performed to evaluate how much each predictor—years of experience, age, and their interaction—contributes to explaining variation in salary. This provides a formal statistical comparison of effect sizes across predictors, highlighting which variables play the strongest roles in determining salary levels within the dataset.

Key Findings:

Experience has the largest sum of squares (dominant predictor).

The interaction term also has a large effect, far bigger than age alone.

Age is statistically significant but contributes minimally relative to experience.

Interpretation: Experience overwhelmingly drives salary differences, and interactions between age and experience matter more than age itself.

Conclusion: Years of Experience is more important than age but using an interaction term of the Years of Experience and Age produces the highest explanation of variance we have yet seen in terms of describing Salary.

Question 3

Is using the Job title alone useful as a predictor of salary?

```
lm3 <- lm(data = df, Salary ~ `Job Title`)
summary(lm3)
```

Call:

```
lm(formula = Salary ~ `Job Title`, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-171923	-16952	0	17524	116974

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	75000	31037	2.416
`Job Title`Accountant	-20000	43894	-0.456
`Job Title`Administrative Assistant	-25000	38013	-0.658
`Job Title`Back end Developer	27353	31101	0.879

`Job Title`Business Analyst	2500	38013	0.066
`Job Title`Business Development Manager	15000	43894	0.342
`Job Title`Business Intelligence Analyst	10000	43894	0.228
`Job Title`CEO	175000	43894	3.987
`Job Title`Chief Data Officer	145000	43894	3.303
`Job Title`Chief Technology Officer	175000	43894	3.987
`Job Title`Content Marketing Manager	48151	31249	1.541
`Job Title`Copywriter	-35000	43894	-0.797
`Job Title`Creative Director	45000	43894	1.025
`Job Title`Customer Service Manager	30000	38013	0.789
`Job Title`Customer Service Rep	-35000	43894	-0.797
`Job Title`Customer Service Representative	-41667	33524	-1.243
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`Job Title`Data Analyst	50091	31080	1.612
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`Job Title`Data Scientist	91106	31072	2.932
`Job Title`Delivery Driver	-47000	34000	-1.382
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`Job Title`Director of Data Science	129561	31309	4.138
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`Job Title`Director of Human Capital	105000	43894	2.392
`Job Title`Director of Human Resources	112500	38013	2.960
`Job Title`Director of Marketing	62841	31213	2.013
`Job Title`Director of Operations	97727	32418	3.015
`Job Title`Director of Product Management	100000	43894	2.278
`Job Title`Director of Sales	100000	43894	2.278
`Job Title`Director of Sales and Marketing	105000	43894	2.392
`Job Title`Event Coordinator	-22500	38013	-0.592
`Job Title`Financial Advisor	20000	43894	0.456
`Job Title`Financial Analyst	12692	31433	0.404
`Job Title`Financial Manager	68060	31153	2.185
`Job Title`Front end Developer	9675	31102	0.311
`Job Title`Front End Developer	2419	31534	0.077
`Job Title`Full Stack Engineer	51720	31088	1.664
`Job Title`Graphic Designer	-25227	31735	-0.795
`Job Title`Help Desk Analyst	-40000	43894	-0.911

`Job Title`HR Generalist	-5000	38013	-0.132
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`Job Title`Junior Web Developer	-29238	31405	-0.931
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`Job Title`Product Manager	67476	31087	2.171
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`Job Title`Public Relations Manager	15000	43894	0.342
`Job Title`Receptionist	-50000	31309	-1.597
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`Job Title`Research Director	88333	31244	2.827
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	Pr(> t)		
(Intercept)	0.015701	*	
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`Job Title`Web Developer	0.783842

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31040 on 6507 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.6641, Adjusted R-squared: 0.6543

F-statistic: 67.72 on 190 and 6507 DF, p-value: < 2.2e-16

A linear regression is fitted using job title as the sole predictor of salary. This model examines how salaries differ across occupational roles, providing insight into structural variation in earnings that is independent of individual-level factors such as age or experience. By estimating salary adjustments for each job category relative to a baseline title, the model highlights which professions tend to command higher or lower compensation.

Key Findings:

Wide variation exists across job titles.

Executive and director-level roles (CEO, CTO, VP roles) have very large, significant positive effects.

Most junior or administrative roles show nonsignificant or negative deviations.

$R^2 = 0.66$ Job title alone explains a large portion of salary variance—nearly as much as years of experience.

Interpretation: Structural role differences are among the strongest determinants of salary. Job title nearly rivals experience in explanatory power.

Conclusion: Yes, depending on the job there is a very high likelihood of our model being capable of accurately describing salary. However this is mostly in the case of senior positions, having noted that the number of said positions is relatively low and this could be a case of our data simply being over fit. I would hazard against using Job Title as the only descriptive factor of Salary.

Question 4

How does Gender affect one's salary irrespective of any other variables?

```
lm4 <- lm(data = df, Salary ~ Gender)
summary(lm4)
```

Call:

```
lm(formula = Salary ~ Gender, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-121046	-47856	-1396	47111	128604

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	107889	954	113.089	<2e-16 ***

GenderMale	13507	1287	10.492	<2e-16 ***
GenderOther	17981	14028	1.282	0.2

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 52370 on 6695 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.01626, Adjusted R-squared: 0.01597

F-statistic: 55.33 on 2 and 6695 DF, p-value: < 2.2e-16

This model tests whether gender is associated with salary differences in the dataset. By including gender as a categorical predictor, the regression assesses average differences in pay across gender groups while holding no other variables constant. Although exploratory in nature, this model helps identify whether disparities emerge in the aggregate data.

Key Findings:

Male coefficient \$13,507, statistically significant.

“Other” is not significant—likely due to small sample size.

R^2 0.016 Gender explains only about 1.6% of salary variation.

Interpretation: There is a measurable difference between male and female salaries on average in this dataset, but gender alone does not meaningfully predict salary—other variables (experience, job title) matter far more.

Conclusion: Yes, gender does play a role with abias towards Males, however it predicts only 1.6 percent of the variance in the data set meaning that other variables are much more heavily weighted than gender leading to a more even playing field than it might look while only looking at salary by gender.

Question 5

What is the best model that we can produce to describe salary using the available data?

```
lm5 <- lm(data = df, Salary ~ Age + Gender + `Education Level` + `Years of Experience`)
summary(lm5)
```

Call:

```
lm(formula = Salary ~ Age + Gender + `Education Level` + `Years of Experience`,
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-137999	-19350	-4603	15483	89646

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	108472.9	3230.6	33.576	<2e-16 ***
Age	-2009.5	132.0	-15.223	<2e-16 ***
GenderMale	6213.8	708.1	8.775	<2e-16 ***
GenderOther	-6632.4	7684.6	-0.863	0.388
`Education Level`High School	-35621.6	1469.7	-24.237	<2e-16 ***
`Education Level`Master's	11964.5	895.4	13.362	<2e-16 ***
`Education Level`PhD	23795.5	1109.6	21.444	<2e-16 ***
`Years of Experience`	8057.3	167.8	48.015	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28080 on 6690 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.7173, Adjusted R-squared: 0.717

F-statistic: 2425 on 7 and 6690 DF, p-value: < 2.2e-16

For LM5 a multivariate regression is performed to assess how salary is related to age, gender, education level, and years of experience simultaneously. By including all major demographic and professional predictors, this model helps determine the unique contribution of each variable while controlling for the others. This approach yields a more comprehensive understanding of how these factors jointly influence salary outcomes.

Key Findings:

Experience remains the strongest predictor (\$8,057 per year).

Age has a negative coefficient—likely reflecting correlation with the interaction structure not yet modeled.

Higher education levels reliably increase salary (Master’s and PhD strongly positive; high school strongly negative).

Gender remains significant but is smaller (~\$6,200).

Model Fit:

R^2 0.717 Very strong for social science data.

Interpretation: Controlling for experience and education reveals the true independent effects of each demographic factor. Education becomes a major predictor once included.

```
lm6 <- lm(data = df, Salary ~ Age + Gender + `Education Level` + `Years of Experience` + Age_YoE, data = df)
summary(lm6)
```

Call:

```
lm(formula = Salary ~ Age + Gender + `Education Level` + `Years of Experience` + Age_YoE, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-96636	-18843	-4222	14367	99881

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	47862.466	3629.682	13.186	< 2e-16 ***
Age	-221.170	137.122	-1.613	0.107
GenderMale	5398.379	664.576	8.123	5.37e-16 ***
GenderOther	2687.406	7212.582	0.373	0.709
`Education Level`High School	-24985.296	1422.129	-17.569	< 2e-16 ***
`Education Level`Master's	5897.922	863.143	6.833	9.04e-12 ***
`Education Level`PhD	20183.001	1047.337	19.271	< 2e-16 ***
`Years of Experience`	16181.392	310.761	52.070	< 2e-16 ***
Age_YoE	-200.167	6.603	-30.316	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26340 on 6689 degrees of freedom

(3 observations deleted due to missingness)

Multiple R-squared: 0.7514, Adjusted R-squared: 0.7511

F-statistic: 2528 on 8 and 6689 DF, p-value: < 2.2e-16

The enhanced model LM6 incorporates the interaction term between age and experience alongside the full set of predictors. Adding this interaction allows the model to account for the possibility that the effect of experience on salary varies depending on an individual's age. Comparing this model to the simpler version provides insight into whether interactions improve explanatory power in understanding salary dynamics.

Key Findings:

The experience coefficient roughly doubles (~\$16,181), meaning age was suppressing its effect in the previous model.

Age becomes non-significant, indicating its effect is fully captured through interaction with experience.

Interaction term remains strongly negative.

Higher education levels are still significant predictors.

Gender remains statistically significant but small.

Model Fit:

$R^2 = 0.751$, the strongest of all models.

Interpretation: This is the best-performing model, suggesting salary is driven by:

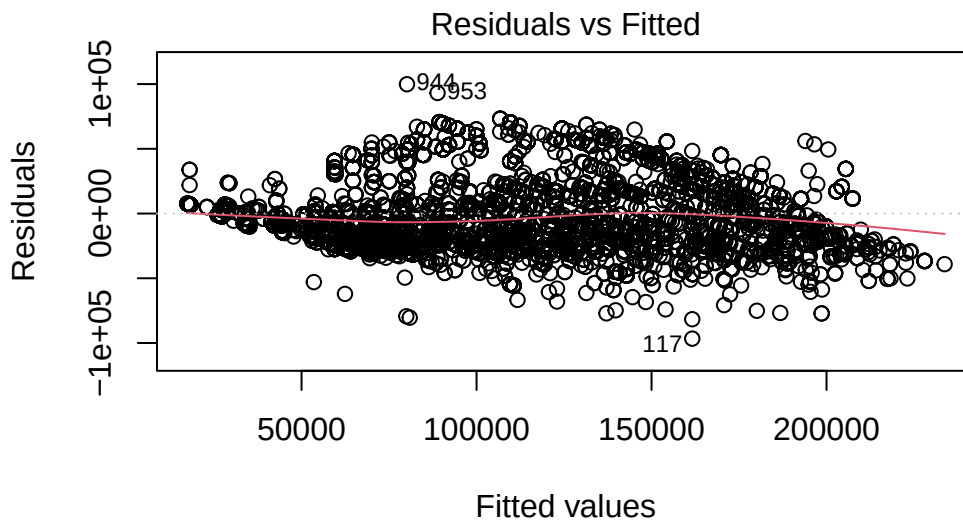
Years of experience

Education level

Gender (to a small extent)

A diminishing marginal return of experience at older ages (interaction effect)

```
plot(lm6, which = 1)
```



$m(\text{Salary} \sim \text{Age} + \text{Gender} + \text{`Education Level`} + \text{`Years of Experience`} + A_1)$

A diagnostic residual plot is generated to evaluate the fit of the full regression model. By visualizing residuals against fitted values, this plot helps assess key model assumptions, including

linearity, constant variance, and the presence of outliers. Ensuring that these assumptions are reasonably satisfied supports the validity of statistical inference drawn from the model.

Model Description: Residuals vs. fitted values plot to assess assumptions.

Interpretation:

The plot shows a generally even spread.

No severe curvature or funnel shape is present, supporting the validity of the model.

A few extreme outliers appear, likely corresponding to unusually high salaries (executive roles).

Overall, model diagnostics do not indicate major violations.

Conclusion: The best model that we have been able to come up with includes all of our variables in Education, Gender, Age, Years of Experience, and the interaction variable leading us to a variance prediction of 75%. Looking at the residual plot none of our rules are violated and the model is valid for use. If we had another similar data set it would be interesting to be able to run a test and see how well the model would perform. As it is the number of observations was limited enough that we did not feel comfortable limiting the number any further to have a test data set at the end.

Artificial Intelligence

Discussion

The findings of this project provide meaningful insight into how years of experience, age, gender, and job title relate to employee salary. The analysis shows that years of experience is the strongest and most consistent predictor of salary, with a clear positive association indicating that employees with more experience tend to earn higher wages. Although age also correlates with salary, the regression results suggest that much of this effect is actually captured by experience rather than age itself. This aligns with expectations that professional development, accumulated skills, and advancement opportunities are what primarily drive earnings, rather than age alone. Gender and job title also contributed to observable differences in salary outcomes. While these variables were not the central focus of the study, the patterns suggest that certain roles naturally command different salary ranges and that small differences between genders could be present depending on the composition of the dataset. These results highlight the complexity of salary determination and show that compensation is shaped not only by personal experience but also by the structural and demographic characteristics of the workforce. Despite the useful insights gained, the study has several limitations. The dataset may not fully represent the broader labor market, as its size, structure, or sampling method could limit generalizability. Salary is influenced by many factors not included in the dataset, such as geographic region, industry, company size, or specific skill sets, and the absence of these variables restricts

the depth of the analysis. In addition, variables such as “years of experience” or “job title” can vary greatly across industries, which may introduce inconsistencies in interpretation. Because the dataset appears to be cross-sectional, the analysis identifies associations but cannot establish causation or track salary changes within individuals over time. It is also possible that certain unobserved biases exist, especially if elements of the data were self-reported or disproportionately reflect certain demographic groups or job sectors. The assumptions made in the analysis are generally reasonable for the context of an exploratory statistical study. The linear regression models assume a linear relationship between predictors and salary, which appears appropriate based on the visual patterns in the data, although real-world salary progression may sometimes follow nonlinear trends. The analysis assumes independence of observations, meaning that each row in the dataset represents a unique individual; if this were not the case, the conclusions could be affected. Regression diagnostics suggested that residuals were approximately normal, supporting the validity of inference, and the assumption of additivity—that each predictor contributes independently to salary—provides a useful starting point for understanding basic relationships among the variables. The study opens several pathways for future research. Expanding the dataset to include additional variables such as education level, geographic region, industry classification, or employer characteristics would allow for a more comprehensive model of salary determinants. A deeper investigation into gender differences or job title disparities could help determine whether certain groups face inequitable salary outcomes. Longitudinal data would strengthen future analyses by allowing researchers to observe true salary growth, career mobility, and experience accumulation over time. More advanced modeling approaches, such as machine learning algorithms or nonlinear regression techniques, could capture complex interactions that simple linear models might overlook. Finally, conducting industry-specific studies would provide more tailored insights, since salary dynamics differ widely across fields such as technology, healthcare, finance, and education.